

$$\begin{bmatrix} 3 & 6 & 1 & 2 \\ 0 & 6 & 3 & 5 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 9 \\ 3 \end{bmatrix}$$

①

$$3x_4 = 3 \rightarrow \underline{x_4 = 1}$$

$$4x_3 + x_4 = 9 \rightarrow 4x_3 + 1 = 9 \rightarrow \underline{x_3 = 2}$$

$$6x_2 + 3x_3 + 5x_4 = -1 \rightarrow 6x_2 + 6 + 5 = -1 \rightarrow \underline{x_2 = -2}$$

$$3x_1 + 6x_2 + x_3 + 2x_4 = 1$$

$$\rightarrow 3x_1 - 12 + 2 + 2 = 1 \rightarrow \underline{x_1 = 3}$$

②

PDF #2

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$$F(x) = 3x^5 - 32x^4 + 140x^3 - 315x^2 + 365x - 174$$

(3)

Upper bound due to Cauchy:

$$|z_i| < 1 + |a_n|^{-1} \cdot \max_{0 \leq i \leq n-1} |a_i|$$

$$\begin{array}{ccc} \downarrow & \downarrow^{-1} & \downarrow (a_1) \\ |z_i| < 1 + |3|^{-1} \cdot 365 \end{array}$$

$$|z_i| < \frac{368}{3} = P_2$$

Lower bound: UB of $S(x) = x^n F(x^{-1})$

$$S(x) = 3 - 32x + 140x^2 - 315x^3 + 365x^4 - 174x^5$$

$$|z_i| < 1 + |a_n|^{-1} \cdot \max_{0 \leq i \leq n-1} |a_i|$$

$$\begin{array}{ccc} \downarrow & \downarrow & \\ |z_i| < 1 + |-174|^{-1} \cdot 365 \end{array}$$

$$|z_i| < \frac{539}{174} = P_1^{-1}$$

$$P_2 = \frac{368}{3} \quad P_1 = \frac{174}{539}$$

x	2	-1	1	0
$P(x)$	-2	-8	2	2

$$L_0(x) = \frac{(x+1)(x-1)x}{(2+1)(2-1)2}$$

$$= \frac{(x+1)(x-1)x}{-6}$$

$$L_1(x) = \frac{(x-2)(x-1)x}{(-1-2)(-1-1)(-1)} = \frac{(x-2)(x-1)x}{-6}$$

$$L_2(x) = \frac{(x-2)(x+1)x}{(1-2)(1+1)1} = \frac{(x-2)(x+1)x}{-2}$$

$$L_3(x) = \frac{(x-2)(x+1)(x-1)}{(0-2)(0+1)(0-1)} = \frac{(x-2)(x+1)(x-1)}{2}$$

$$P_3(x) = -2L_0(x) - 8L_1(x) + 2L_2(x) + 2L_3(x)$$



$$L_i(x) = \begin{cases} 1 & x = x_i \\ 0 & x = x_j, j \neq i \end{cases} \quad (4)$$



$$L_i(x) = \prod_{j=0, j \neq i}^n \frac{(x - x_j)}{(x_i - x_j)}$$

